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Selection Committee
Faculty of Engineering
The University of Sydney
Darlington Campus, NSW 2006, Australia

Dear Members of the Selection Committee,

Application for Lecturer (Level B), Multiple Continuing (Tenure-Track) Academic Positions, Faculty of Engineering, Job 0157992

I am writing to apply for a **Lecturer (Level B)** position in the Faculty of Engineering, with the **School of Electrical and Computer Engineering** as my first preference. My research develops **physics-informed AI for the design and validation of electrical machines and drives**, the energy converters at the centre of electric transport, renewable generation and robotics. I am a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham, on the EPSRC-funded AIMD programme, following a PhD in Electrical Engineering at Southeast University in 2025. My record comprises **24 journal papers (20 in IEEE Transactions, 11 as first author)**, 19 conference papers and 9 granted patents, recognised by the 2025 *IEEE Trans. Transportation Electrification* Prize Paper Award, 1st Prize in the 2025 IEEE TEC Electric Vehicle Design Competition, and the 2025 MDPI *Machines* Best PhD Thesis Award.

My work connects with all three directions in the Faculty's call. To **build AI**, I developed a deep-learning surrogate that predicts the full magnetic field of a machine, reaching $R^2 = 0.9996$ for torque on unseen designs with more than 94% less evaluation time than finite element analysis, and trained it on an **in-house Python finite element package** I wrote. To **leverage AI**, I combined fast models with FEA to cut a 67-hour design optimisation to 2.3 hours, and built thermal and mechanical models validated against thermocouple data and a 22,500 rpm overspeed test. To **understand AI**, I lead a project supported by the OpenAI Researcher Access Program on LLM agents that set up and run engineering simulations, and I am now studying where their answers break physics or disagree with measurement.

What I would add to the Faculty is **validation against hardware**. As technical lead with Inovance, I took two high-speed traction machines from specification through drawing release and supplier manufacture to dynamometer test (20,000 rpm, 270 kW, 97.4% peak efficiency, all measured), and a further IEEE Transactions paper on the design and performance evaluation of hairpin-winding machines was accepted this month. At Sydney I would build a programme on AI models that engineers can trust, starting with an **ARC DECRA** and moving to **ARC Linkage** and CRC-P projects with industry in electrified mining, heavy transport and renewable energy, alongside the School's work on future energy networks, the Net Zero Institute and the Australian Centre for Robotics. My funding record is at an early stage. This year I received a **competitively awarded N-ZEEE Flexible Funding grant (£2,700)** to present my work at IEEE ICEMS 2026 in Singapore, which also gives me a first opportunity to build links in the Asia-Pacific machines and drives community.

In teaching, I was teaching assistant for the Southeast University postgraduate course *Integrated Practice of Electrical Machine Systems*, and I am ready to coordinate units in **electrical machines and drives, power electronics and energy conversion**. My teaching statement sets out how I would organise such a unit around the comparison of model, simulation and measurement, and how I would use AI in it: students train and test a small surrogate themselves, declare their use of AI tools, and are assessed through oral defences and laboratory work that test understanding. I would take the University's teaching development programme and Advance HE Fellowship as an early priority.

My CV, research statement and teaching statement are attached. I would be glad to discuss how I could contribute to the School and the Faculty.

Yours sincerely,

Wentao Zhang